



उत्पाद मैनुअल
आई एस 17293 : 2020 के अनुसार
रेटेड वोल्टेज 1500 डी. सी. की फोटोवोल्टिक प्रणालियों के लिए इलेक्ट्रिक केबल के लिए
दस्तावेज संख्या – पीएम/आईएस 17293/2/ जून 2023

भारतीय मानक ब्यूरो की स्कीम-I (अनुरूपता मूल्यांकन) विनियम, 2018 के तहत यह उत्पाद मैनुअल प्रमाणीकरण के प्रचालन में रीति और पारिश्रिता की सुसंगतता सुनिश्चित करने के लिए सभी क्षेत्रीय/शाखा कार्यालयों और लाइसेंसी द्वारा संदर्भ सामग्री के रूप में उपयोग किया जाएगा। बीआईएस प्रमाणीकरण लाइसेंस/प्रमाणपत्र प्राप्त करने के इच्छुक भावी आवेदकों द्वारा भी इस दस्तावेज का उपयोग किया जा सकता है।

PRODUCT MANUAL FOR
Electric Cables for Photovoltaic Systems for Rated Voltage 1500 V d.c.
ACCORDING TO IS 17293 : 2020
Document No - PM/IS 17293/2/ June 2023

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate.

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PRODUCT MANUAL FOR
Electric Cables for Photovoltaic Systems for Rated Voltage 1500 V d.c.
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1.	Product	:	IS 17293 : 2020
	Title	:	Electric Cables for Photovoltaic Systems for Rated Voltage 1500 V d.c.
	No. of Amendments	:	Nil
2.	Sampling Guidelines:		
a)	Raw material	:	Annealed tinned Copper wire as per IS 8130
b)	Grouping guidelines	:	Please refer Annex – A
c)	Sampling Guidelines During Operation of Licence	:	- One FS shall be drawn once in five year for complete testing of the product* - One FS (except every fifth year) shall be drawn once in a year and tested for all test except – a. Thermal endurance Test – Annex – B & Table 11 b. Damp Heat, Smoke emission of complete cable, Weathering / UV resistance on sheath, Ozone resistance on complete cable, Assessment of Halogens (Insulation & Sheath) – Cl. 11.1 & Table 3 - The other FS shall be drawn annually for Acceptance tests only - Two Market samples to be drawn annually and tested for Acceptance tests only.
3.	Sample Size	:	100 m
4.	List of Test Equipment	:	Please refer Annex – C
5.	Scheme of Inspection and Testing	:	Please refer Annex – D
6.	Possible tests in a day	:	Please refer Annex – E
7.	Scope of the Licence	:	
	Licence is granted to use Standard Mark as per IS 17293:2020 with the following scope:		
	Name of the product	:	Electric Cables for Photovoltaic Systems for Rated Voltage 1500 V d.c.

	Nominal Cross – Sectional area	:	sqmm to upto and including sqmm
	Conductor Class	:	Class 5 / Class 2

*- Report of the Factory Sample drawn for complete testing by BIS is to be shared with the manufacturer. For this variety, manufacturer may not be insisted for complete testing as per SIT. However, manufacturer shall carry out complete testing for other varieties as per SIT.

ANNEX A

Grouping Guidelines

1. Electric Cables for Photovoltaic Systems for Rated Voltage 1500 V d.c. as per IS 17293:2020 are classified based on the following:
 - a. Cross – section area
 - b. Conductor Class
2. Samples of Cable for each conductor class to be tested for GoL / CSoL.
3. When a cable with higher cross sectional area is tested, all the cable with lower cross sectional may be covered in scope of licence, subject to the condition that class of conductor, insulation and sheath materials remain same.
4. Further, in addition to the sizes tested as per the above grouping guidelines, factory test report / independent test report of all other sizes intended to be covered in scope shall also be submitted, subject to the condition that class of conductor, insulation type and sheath type remain same.
5. The Firm shall declare the varieties and sizes of Cables they intend to cover in the Licence. The Scope of Licence may be restricted based on the manufacturing and testing capabilities of the Manufacturer.
6. During the operation of the Licence, BO shall ensure that all the varieties covered in the Licence are tested in rotation to the extent possible.

ANNEX - B
List of Test Equipment

Major test equipment required to test as per the Indian Standard

Sl. No.	Test Equipment	Tests used in with Clause Reference	
		Cl. Ref.	Tests
1	Vernier Callipers	5.3, 7, 11.1	Measurement of thickness & Outer Dia.
2	Steel Scale	11.1	Elongation, flammability
3	Micrometer	11.1	Measurement of Wire Diameter.
4	Measuring Microscope	5.3, 7, 11.1	Measurement of thickness (insulation & sheath) and hot deformation test
5	Graduated Magnifying Glass	5.3, 7, 11.1	Measurement of thickness (insulation & sheath) and hot deformation test
6	Tensile Testing Machine	11.1	Tensile Strength, Elongation Test & Annealing Test
7	Dumb-Bell Cutting Machine with Die	11.1	Tensile Strength & Elongation Test
8	Hot Air oven with thermostatic Temperature controller	11.1	Heat Shock, Hot Deformation & Shrinkage Test
9	Thermal Ageing Oven with Thermostatic Temperature controller, Air flow Meter & Hour Meter	11.1	Tensile Strength & Elongation After ageing, Loss Of Mass Test
10	Water Bath with thermostatic temperature controller, Stirrer & Hour Meter	11.1, 11.2, 11.4	Water immersion A.C. & D.C. Test, Insulation Resistance Constant/ Volume Resistivity Test
11	Mega ohmmeter	6.3, 7, 11.1, 11.2.3, 11.3.3	Insulation Resistance Constant/ Volume Resistivity Test
12	Smooth metal Mandrels	11.1	Cold Bend Test
13	Low temperature Impact Apparatus	11.1	Cold Impact Test
14	Hot Pressure Test Apparatus with Weights	11.1	Hot Deformation Test
15	Physical Weighing Balance with Weights	11.1	Persulphate Test, Flammability Test

16	Kelvin Double Bridge with galvanometer, D.C. Source	11.1	Conductor Resistance Test
17	Flammability Test apparatus with burner, Gas cylinder, Scale, Stopwatch & 0.71 mm copper wire	11.1	Flammability Test
18	Conditioning chamber & Refrigerator with Humidity Indicator	11.1	Tensile Strength, Elongation Test, Cold Bend, Cold Impact & Thermal stability Test
19	AC Spark Tester	11.2.2	Spark Testing
20	D.C. High Voltage Test set	11.1	D.C. High Voltage Test
21	Glass Thermometer	6.3, 7, 11.1, 11.2.3, 11.3.3	Conductor Resistance & Insulation Resistance Test
22	Balance Digital	11.1	Flammability Test
23	Stop Watch	11.1	Flammability Test
24	Test setup for Ozone Resistance Test	11.1, Annex G	Ozone
25	Tube Furnace, Quartz Combustion Tube, Porcelain combustion boat, Wash Bottles, Glass Tubing and Silicon Rubber Stoppers, Air Flow Meter, Reagents	11.1, Annex J	Halogen Acid Gas Evolution Testing
26	Smoke density test apparatus	11.1, Annex A	Smoke density rating test

The above list is indicative only and may not be treated as exhaustive.

ANNEX C**Scheme of Inspection and Testing**

- 1. LABORATORY** - A laboratory shall be maintained which shall be suitably equipped (as per the requirement given in column 2 of Table 1) and staffed, where different tests given in the specification shall be carried out in accordance with the methods given in the specification.

1.1 The manufacturer shall prepare a calibration plan for the test equipment.

- 2. TEST RECORDS** – The manufacturer shall maintain test records for the tests carried out to establish conformity.

- 3. LABELLING AND MARKING** – As per Cl. 8 of IS 17293:2020. The Standard Mark as given in the Schedule of the licence shall be either printed or indented or embossed on the cable.

3.1 Information as per Cl. 12.2 of IS 17293:2020 shall also be either stenciled on reel/drum or contained on the label attached to the coil.

3.2 In addition to the above, the licence no. CM/L-_____ and details of BIS website shall be marked either on the reel/drums or on the label attached to the coil as follows: “For details of BIS certification please visit www.bis.gov.in”

- 4. CONTROL UNIT** – Cable of each nominal cross-sectional area and class of conductor manufactured under similar conditions in a day shall constitute a control unit.
- 5. LEVELS OF CONTROL** - The tests as indicated in column 1 of [Table 1](#) and the levels of control in column 3 of [Table 1](#), shall be carried out on the whole production of the factory which is covered by this plan and appropriate records maintained in accordance with paragraph 2 above.
- 6. REJECTIONS** – Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

TABLE 1

1				2	3		
Test Details				Test equipment requirement	Levels of Control		
		Test Methods		R: required (or) S: Sub-contracting permitted			
Cl.	Requirement	Clause	Ref		No. of Sample	Frequency	Remarks
4.1	Conductor		IS 8130	S	One	Each Consignment	No further testing is required if accompanied with Test Certificate or with BIS Standard mark
11	Requirements for Completed Cables						
4	Conductor						
Table 3	Resistance	Part 5	IS 10810	R	Firm to have adequate in-process controls to check compliance of this parameter as per the tolerances given in the Indian Standard. However, appropriate records shall be maintained by the manufacturer for evidence of conformity		
	Maximum diameter of wire in conductor	Part 4	IS 10810	R			
	Persulphate test	Method B of Part 4	IS 10810	R	One	Each Control Unit	
5	Insulation						
5.2	Insulation Application	5.2	IS 17293	R	One	Each Control Unit	

						Each Control Unit	
5.3	Insulation Thickness	5.3	IS 17293	R	One		
Annex – B & Table 11 (Cl. 5.1 & 6.1)	Tensile strength before ageing	Part 7	IS 10810	R	One		
	Elongation at break before ageing	Part 7	IS 10810	R	One		
	Tensile strength after ageing	Part 11	IS 10810	R	One	Cable of each size and manufactured in a year from same type of cross linked compound	If there is a change in material/manufacturing process, then additional tests has to be carried out
	Elongation at break after ageing				One		
	Hot set test (Insulation)	Part 30	IS 10810	R	One		
	Bending at low temperature	Part 20	IS 10810	S	One		
	Elongation at low temperature	Annex F	IS 17293	S	One		
	Compatibility test	Part 11	IS 10810	S	One		
	6	Sheath					
6.2	Sheath application	6.2	IS 17293	R	One	Each Control Unit	
6.3	Sheath Thickness	6.3	IS 17293	R	One		
6.4 11.3.1	Sheath Colour	6.4	IS 17293	R	One		
Annex – B & Table 11 (Cl. 5.1	Tensile strength before ageing	Part 7	IS 10810	R	One		
	Elongation at break before ageing	Part 7	IS 10810	R	One		

& 6.1)	Tensile strength after ageing	Part 11	IS 10810	R	One	Cable of each size manufactured in a year from same type of cross linked compound	If there is a change in material/manufacturing process, then additional tests has to be carried out
	Elongation at break after ageing				One		
	Bending at low temperature	Part 20	IS 10810	S	One		
	Elongation at low temperature	Annex F	IS 17293	S	One		
	Compatibility test	Part 11	IS 10810	S	One		
7	Overall Dimension	7	IS 17293	R	Firm to have adequate in-process controls to check compliance of this parameter as per the tolerances given in the Indian Standard. However, appropriate records shall be maintained by the manufacturer for evidence of conformity		
8 11.3.2	Cable Identification	8	IS 17293	R			
8.1	Continuity of Marking	8.1	IS 17293	R			
8.2	Durability	8.2	IS 17293	R			
8.3	Legibility	8.3	IS 17293	R			
9	Core Identification	9	IS 17293	R			
12	Packing and Marking	12.1, 12.2, 12.3	IS 17293	R			
11.2	Electrical Tests						
11.3.3	Overall Diameters & Ovality	11.3.3 & Table 1	IS 17293	R	Firm to have adequate in-process controls to check compliance of this parameter as per the tolerances given in the Indian Standard. However, appropriate records shall be maintained by the manufacturer for evidence of conformity		
Table 3	Test for vertical flame propagation on complete cables	Part 53	IS 10810	R	One	Each Control Unit	
11.2.1	Voltage test on completed cable	11.2.1	IS 17293	R	One		
11.2.2	Spark Test	Part 44	IS 10810	R	One		

11.2.3	Measurement of Insulation Resistance	Table 1	IS 17293	R	One	Each Control Unit	
11.2.4	Long-term Resistance of Insulation d.c.	11.2.4	IS 17293	S	One	Cable of each size and each class of conductor manufactured in a year provided there is no change in material and manufacturing process	If there is a change in material/manufacturing process, then additional tests has to be carried out
11.2.5	Surface Resistance of Sheath	11.2.4	IS 17293	S	One		
Cl. 11.1 & Table 3	Cold impact test	Part 21	IS 10810	S	One		
	Cold bend test	Part 20	IS 10810	S	One		
	Ozone resistance on complete cable	Annex G	IS 17293	S	One		
Cl. 11.1 & Table 3	Weathering / UV resistance on sheath	Annex E	IS 17293	S	One		
	Dynamic penetration test	Annex D	IS 17293	S	One		
	Damp heat test	Annex H	IS 17293	S	One		
	Shrinkage test on sheath	Part 12	IS 10810	S	One		
	Smoke emission of complete cable	Part 63	IS 10810	S	One		
	Assessment of Halogens	Annex J	IS 17293	S	One		
Annex – B & Table 11 (Cl. 5.1 & 6.1)	Thermal endurance Test		IS 8504 (Part 1) and (Part 2)	S	One sample of each class of conductor for any size manufactured in five years provided there is no change in material and manufacturing process (Refer Note-3)		

Note-1: Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empaneled by the Bureau.

Note-2: Levels of control given in column 3 are only recommendatory in nature. The manufacturer may define the control unit/batch/lot and submit his own levels of control in column 3 with proper justification for approval by BO Head.

Note-3: Report of the Factory Sample drawn for complete testing by BIS is to be shared with the manufacturer. For this variety, manufacturer may not be insisted for complete testing as per SIT. However, manufacturer shall carry out complete testing for other varieties as per SIT.

ANNEX D

Possible tests in a day

1. Dimensions
 - a. Maximum Conductor diameter
 - b. Overall dimensions
 - c. Ovality
 - d. Thickness of sheath and insulation
 - e. Insulation Application
 - f. Insulation Thickness
 - g. Sheath application
 - h. Sheath Thickness
 - i. Sheath Colour
 - j. Cable Identification
 - k. Continuity of Marking
 - l. Durability
 - m. Legibility
 - n. Core Identification
2. Persulphate test
3. Conductor Resistance
4. Tensile Test (before ageing)
5. Insulation Resistance Test
6. HV Test (at Room Temperature)
7. Flammability Test